

NETWORKS AND SWITCHING

LECTURE 4 - ASSIGNMENTS

PART 1 – Convert between bases

Question 1. Convert the following numbers from the base shown to base 10

- a) 111 (base 2) = $1+2+4=7$
- b) 777 (base 8) = 511
- c) FEC (base 16) = 4076

Question 2. Convert the following binary numbers to hexadecimal.

- a) 10101001 169 = A9
- b) 11100111 231 = E7
- c) 01101110 110 = 6E
- d) 01121111

Question 3. Convert the following decimal numbers to octal.

- a) 901 = 1605
- b) 321 = 501

Question 4. Convert the following decimal numbers to binary

- a) 45 = 00101101
- b) 69 = 01000101

Question 5. Convert the following decimal numbers to hexadecimal.

- a) 1066 = 42A
- b) 1939 = 793

i will use !@#%&*+

Question 6. If you were going to represent numbers in base 18, what symbols might you use to represent the decimal numbers 10 through 17 other than letters?

Question 7. What is the main difference between the ASCII and Unicode character sets?

PART 2. IPv4

11011111.00010010.00101110.00011011

Question 1. What is the 32-bit binary equivalent of the IP address 223.18.46.27?

Question 2. What is the 32-bit binary equivalent of the IP address 192.168.1.14

Question 3 11000000.10101000.00000001.00001110

What are the classes of the following IP addresses. Please identify the netid and hostid parts of each IP address

- a) 10.162.8.2/10 netbit=10, hostbit=22 subnet 255.192.0.0

- b) 192.65.248.132/26 netbit=26, hostbit= 6 subnet 255.255.255.192
- c) 214.170.185.46/31 netbit=31, hostbit= 1 subnet 255.255.255.254
- d) 129.163.4.161/22 netbit=22, hostbit= 10 subnet 255.255.252.0
- e) 257.45.15.12/30 netbit=30, hostbit= 2 subnet 255.255.255.252